GGG-P-446a
February 17, 1965
SUPERSEDING
Fed. Spec. GGG-P-446
MAY 10, 1954

FEDERAL SPECIFICATION

PLATE, LAPPING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers one type of lapping plate.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 **Classification.** Lapping plates covered by this specification shall be one type and shall be furnished in the following sizes, as specified (see 6.2).

Size 2 by 4 inches.
Size 3 by 5 inches.
Size 6 by 6 inches.
Size 8 by 8 inches.
Size 10 by 14 inches.
Size 14 by 18 inches.
Size 18 by 24 inches.
Size 24 by 48 inches.
Size 30 by 84 inches.
Size 36 by 72 inches.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated Plywood.

PPP-B-621—Boxes, Wood Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-650—Box, Fiberboard, Special Purpose.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are avail-

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able without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–H–15424—Hand Tools; Packaging of.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Official Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

American Standards Association (ASA) Standard:

B46.1—Surface Texture.

(Application for copies should be addressed to American Standards Association, 10 East 40th Street, New York 16, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of lapping plates otherwise conforming to this specification.

3.2 **Construction.** Construction shall be free from any characteristics or defects that may render the lapping plate unsuitable or inefficient for the purpose intended.

3.3 **Design.** The lapping plates shall be the manufacturer's latest design. All parts shall be new, designed and constructed to the requirements specified herein. Lapping plates 8 by 8 inches and smaller shall be solid, while sizes greater than 8 by 8 inches shall include diagonal cross ribbing or other suitable means to insure maximum stability. Three point bearing support shall be provided at the junction of such ribbing. Lapping plates 14 by 18 inches and larger shall be provided with handles, lifting holes, or other suitable means for handling.

3.4 **Materials.** The plates covered by this specification shall be made of fine grain gray iron or iron alloy castings.

3.4.1 *Castings.* All castings shall be of uniform quality, free from defects, scale, and mismatching. In no event shall such processes as welding, peening, or filling with cold solders or metallic pastes be used for reclaiming any casting.

3.4.1.1 *Seasoning.* All castings used in the fabrication of the lapping plates covered by this specification after rough machining shall be stress relieved before final finishing. Stress relieving procedures shall be the manufacturer's standard which will produce lapping plates of the accuracy specified herein.

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3.5 Dimensions and weights. Lapping plates furnished under this specification shall conform to the dimensions and weights as listed in table I for the size specified (see 6.2).

3.6 Hardness. The work surface of lapping plates shall have a Brinell hardness of not less than 180 when tested as specified in 4.4.1.

3.7 Working surfaces. The working surfaces of all lapping plates shall be machine finished to the accuracies and surface finish specified herein.

3.7.1 *Grooves.* Unless otherwise specified (see 6.2), the working surface of the lapping plate shall be grooved to form a square or diamond pattern with the grooves either "V" shaped or straight sided. Plates 14 by 18 inches and over shall have grooves spaced approximately ½ inch center to center and be approximately 3/32 inch wide by 1/16 inch deep. Plates 10 by 14 inches shall have grooves spaced approximately ⅜ inch center to center and approximately 1/16 inch wide by 3/64 inch deep. Plates 8 by 8 inches and less shall have grooves approximately ⅛ inch center to center and approximately 1/32 inch wide by 1/32 inch deep. For the diamond pattern, the smaller angle between the grooves shall be from 45 to 75 degrees (see figures 1 and 2).

TABLE I. *Dimensions and weights*

Size	Working surface dimensions (approximate)		Top thickness (minimum)	Depth overall (minimum)	Weight (minimum)
Inches	*Width, inches*	*Length, inches*	*Inches*	*Inches*	*Pounds*
2 by 4	2	4	5/8	5/8	1-1/2
3 by 5	3	5	3/4	3/4	2-3/4
6 by 6	6	6	7/8	7/8	8
8 by 8	8	8	7/8	7/8	10
10 by 14	10	14	3/4	4	85
14 by 18	14	18	3/4	4	70
18 by 24	18	24	3/4	4	150
24 by 48	24	48	3/4	4	400
30 by 84	30	84	1	6	1,300
36 by 72	36	72	1	6	1,400

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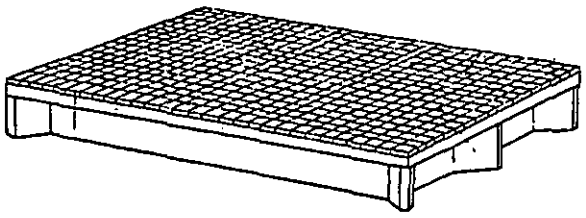


FIGURE 1. *Square pattern.*

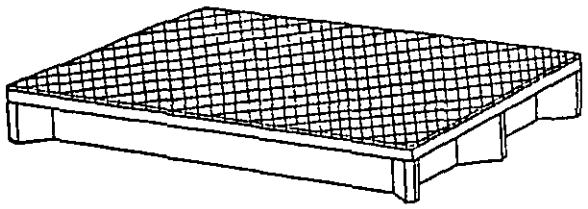


FIGURE 2. *Diamond pattern.*

3.8 Accuracy. The accuracy of the working surface shall be such that no two points on that surface vary with each other by more than 0.0002 inch per linear foot of space separating the two points when tested as specified in 4.4.2.

3.9 Surface roughness. The maximum surface roughness of the working surface shall not exceed 32 microinches arithmetic average (AA) in accordance with ASA B46.1. The rest pads shall be machined and all edges rounded or beveled. The four sides of the top shall be machine finished to 125 microinches AA or better.

3.10 Painting. All surfaces not machined shall be given one coat of primer and two coats of lacquer or enamel. Unless otherwise specified (see 6.2), the color shall be the manufacturer's standard.

3.11 Marking. Lapping plates shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.12 Covers. All lapping plates other than 3 by 5, 6 by 6, and 8 by 8 inch size (see 3.13) shall be furnished with a properly fitted wood cover so designed as to protect the working surface as well as the edges. Each cover shall be given at least two coats of shellac.

3.13 Wood case. Wood cases shall be furnished for the 2 by 4, 3 by 5, 6 by 6, and 8 by 8 inch size lapping plates. The wood cases shall be properly fitted and substantially constructed of hardwood to provide complete protection, and shall be suitably coated with shellac or similar material. When ordered in sets of 2 or 3 (see 6.2), the case shall be provided with separate compartment for each plate (see figure 3).

3.14 Workmanship. The workmanship of the lapping plates shall be of a quality prevailing among manufacturers normally producing lapping plates of the type and sizes specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 *Lot.* All lapping plates of the same size and design presented at the same time

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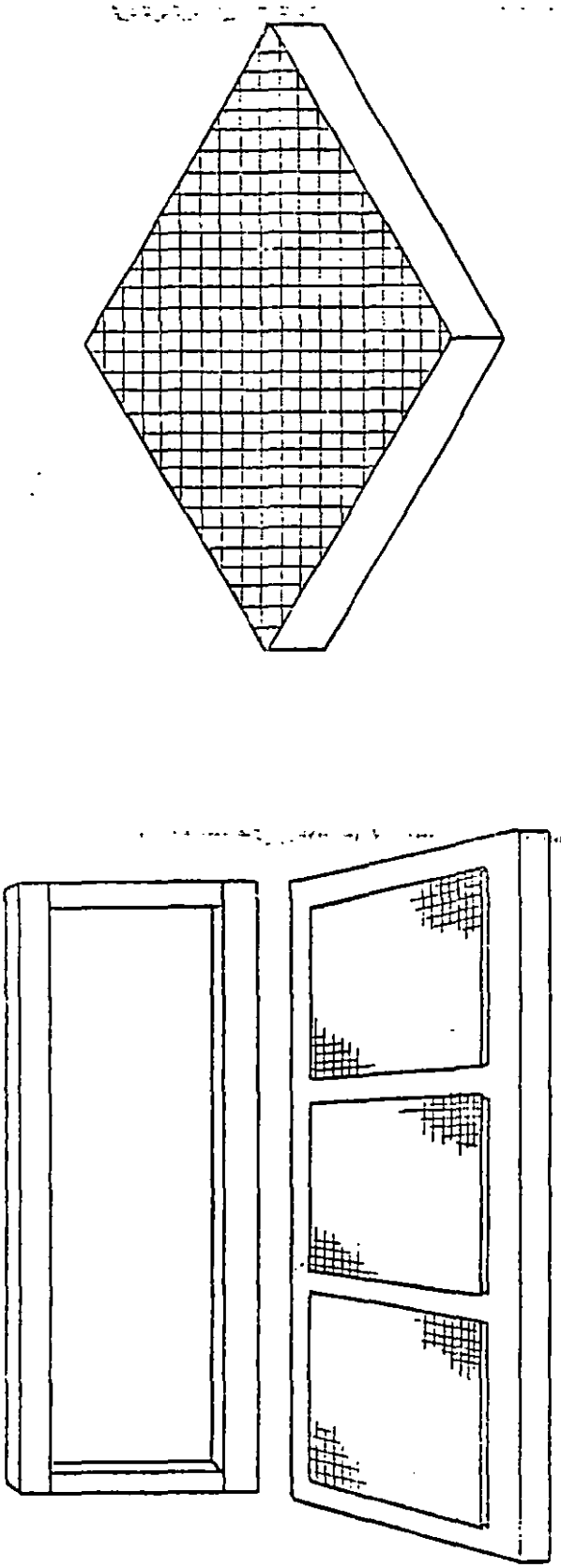


FIGURE 3. *Case with set of 3 blocks of small size lapping plates.*

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shall be considered a lot for purposes of inspection.

4.2.2 *Sampling for quality conformance inspection.* A random sample of lapping plates shall be selected for each lot offered for quality conformance inspection in accordance with MIL–STD–105, inspection level II.

4.3 **Quality conformance inspection** Each of the sample lapping plates selected in accordance with 4.2.2 shall be inspected to verify compliance with this specification. Inspection quality conformance shall be as follows:

(a) Examination (see 4.3.1).

(b) Tests (see 4.3.2 and 4.4).

(c) Examination of preservation, packaging, packing, and marking (see 4.6).

4.3.1 *Examination.* Each of the lapping plates selected in accordance with 4.2.2 shall be examined for conformance with the requirements of this specification. Examination shall be conducted as specified in table II. The acceptable quality level (AQL) shall be as indicated in table II. Any lapping plates in the sample containing one or more defects shall not be offered for delivery, and if the number of defective lapping plates in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

TABLE II. *Classification of defects in accordance with MIL–STD–105*

Categories	Defects
Critical:	None defined.
Major:	AQL = 1.5 percent defective.
101	Size and pattern of lapping plate not as specified.
102	Evidence of cracks, blow holes, porosity, hard spots, shrinkage, scale, or molding sand; casting not of uniform quality.
103	Casting has been plugged, peened, welded, or evidence of unauthorized repairs.
104	Working surface not free of rust.
105	Grooves missing, incomplete, irregular, or not "V" shaped (or straight sided).
106	Spacing between grooves not as required.
107	Width or length not within the specified tolerance.
108	Thickness of top less than the specified minimum.
Minor	AQL = 2.5 percent defective.
201	Width or depth of grooves not as specified.
202	Sides of working surface not machine finished.
203	Edges of rest pads not rounded or beveled.
204	Painting missing defective, incomplete, or not as required.
205	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.
206	Cover or case missing, defective, not properly fitted, not coated, or not as specified.

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4.3.2 *Tests.* Each of the sample lapping plates selected in accordance with 4.2.2 shall be tested in accordance with 4.4 to determine compliance with the requirements of this specification. The AQL shall be 1.5 percent defective. Any plate failing the test shall not be offered for delivery and if the number of nonconforming lapping plates in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 *Hardness test.* Hardness of the lapping plate shall be determined in conformance with method 242 of Fed. Std. No. 151 using a 3,000-kilogram load and a 10 millimeter diameter ball. Readings shall be taken on sections adjacent to the working surface. Preparation of test surfaces shall be made in accordance with method 242 of Fed. Std. No. 151. Failure of the plate to meet the minimum requirements of 180 Brinnel hardness number shall be cause for rejection.

4.4.2 *Accuracy test of working surfaces.* Accuracy of the working surfaces shall be determined by checking flatness of the working surface for conformance with the tolerances specified herein (see 3.8). Methods used in determining conformance shall be an approved method, standard with industry and acceptable to the procuring agency. Failure to meet the accuracy requirements specified shall be cause for rejection.

4.4.3 *Surface roughness test.* The entire working surface shall be checked to determine smoothness of the working surface. The finished working surface shall not exceed a surface roughness of 32 microinches AA center line average (CLA) in accordance with ASA B46.1. Sides of the top shall not exceed 125 microinches AA. Failure to comply with these requirements shall be cause for rejection.

4.4.4 *Weight test.* Each sample lapping plate shall be weighed to determine compliance with the minimum weight requirements of table I for the size being tested. Failure to meet the minimum weight requirements shall be cause for rejection.

4.5 Reports. When specified (see 6.2), the contractor shall furnish reports on tests, in duplicate, showing the results for all tests and analysis of materials used.

4.6 Examination of the preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL–STD–105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE III. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Preservation and packaging	Level not as specified; method not as specified.
Packing	Level not as specified. Contents not blocked, braced, and cushioned. Containers not as specified.

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TABLE III. *Classification of preparation for delivery defects* (continued)

Examine	Defects
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.
Materials	Any component missing or damaged.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of preservation, packaging, and packing shall be in accordance with Fed. Std. No. 102.

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified, (see 6.2).

5.1.1 *Level A.* Each lapping plate shall be preserved and packaged in accordance with MIL–P–116, method I.

5.1.2 *Level B.* Unless otherwise specified (see 6.2), each lapping plate shall be preserved and packaged as specified in 5.1.1.

5.1.3 *Level C.* The lapping plates shall be preserved and packaged in accordance with industry practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Lapping plates, of one size and description, preserved and packaged as specified in 5.1, shall be packed in any of the following containers. Selection of the container or containers shall be at the option of the contractor.

Specification	*Type of class*
PPP–B–585	Class 3
PPP–B–601	Overseas type
PPP–B–621	Class 2
PPP–B–636	Weather-resistant
PPP–B–650	Type (as applicable)

Contents shall be blocked, braced, and cushioned, as required, to prevent damage dur-during handling, shipment, and storage. The gross weight of each container shall not exceed the weight limitations of the applicable container specification. The containers shall be closed, strapped, or banded in accordance with the applicable container specification or appendix thereto.

5.2.2 *Level B.* Lapping plates, of one size and description, preserved and packaged as specified in 5.1, shall be packed in any of the following containers. Selection of the container or containers shall be at the option of the contractor.

Specification	*Type of class*
PPP–B–585	Class 1
PPP–B–601	Domestic type
PPP–B–621	Class 1
PPP–B–636	Class — domestic
PPP–B–650	Type (as applicable)

Contents shall be blocked, braced, and cushioned, as required, to prevent damage during handling, shipment, and storage. The gross weight of each container shall not exceed the weight limitations of the applicable container specification. The containers shall be closed, strapped, or banded in ac-

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cordance with the applicable container specification or appendix thereto.

5.2.3 *Level C.* The lapping plates shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order (see 6.2), the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order (see 6.2), the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lapping plates covered by this specification are intended to be used in lapping operations in ordinary machine shop practices to assure a perfect fit of matching parts. Smaller sizes are used in diesel engine maintenance operations in lapping precision fitted parts and are often used in sets of 2 or 3.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Size required (see 1.2 and table I).

(*c*) Square or diamond pattern, if particular pattern is desired (see 3.7.1).

(*d*) V-shaped or straight sided, if a particular groove is desired (see 3.7.1).

(*e*) If other than manufacturer's standard color is desired (see 3.10).

(*f*) If 2 by 4, 3 by 5, 6 by 6, or 8 by 8 are ordered in sets of 2 or 3, so state the type of case required (see 3.13).

(*g*) Reports, if required (see 4.5).

(*h*) Preservation, packaging, and packing, if different (see 5.1).

(*i*) Level of preservation, packaging, and packing required (see 5.1).

(*j*) Special marking, if required (see 5.3).

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Plates, lapping, steel.

Carload minimum weight 30,000 pounds.

Motor:

Plates, lapping, steel.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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